DO NOW · AO1 REVISION

Mains electricity and the National Grid

Electricity
Paper 1 · Combined and Separate
32 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 15 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**What do a.c. and d.c. stand for?**	Alternating current and direct current	☐
2 ★	**What is an alternating current (a.c.)?**	A current that repeatedly reverses direction.	☐
3 ★	**State the frequency and potential difference of the UK mains electricity supply.**	50 Hz and about 230 V	☐
4 ★	**State the colour of the insulation on the live, neutral and earth wires.**	Live: brown Neutral: blue Earth: green and yellow stripes	☐
5 ★	**State the function of the live wire in a three-core cable.**	It carries the alternating potential difference from the supply.	☐
6 ★	**State the function of the neutral wire in a three-core cable.**	It completes the circuit.	☐
7 ★	**State the function of the earth wire in a three-core cable.**	It is a safety wire that stops the appliance becoming live.	☐
8	**State the potential difference between the live wire and the earth wire.**	About 230 V	☐
9 ★	**Name the wires in a three-core cable that are at, or close to, 0 V.**	The earth wire and the neutral wire	☐
10	**State when the earth wire will carry a current.**	Only when there is a fault.	☐
11 ★	**In a plug, name the wire that is connected to the fuse.**	The live wire	☐
12 ★	**Explain why touching the live wire is dangerous even when the appliance is switched off.**	The live wire is still at about 230 V, so a current would flow through the person to earth.	☐

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	QUESTION	ANSWER	✓
13	**Define power in terms of energy.**	The rate at which energy is transferred.	☐
14 ★	**State the standard unit of power.**	Watt (W)	☐
15	**Give two things that affect the energy transferred by an appliance.**	Its power and the time it is switched on	☐
16	**What does a higher power rating tell you about an appliance?**	It transfers more energy each second.	☐
17 ★	**Describe the National Grid and its purpose.**	A system of cables and transformers linking power stations to consumers.	☐
18	**State what is transferred across the National Grid.**	Electrical power	☐
19	**State the function of transformers.**	To increase or decrease the potential difference.	☐
20	**What does a step-up transformer do to the potential difference?**	It increases it.	☐
21 ★	**What does a step-up transformer do to the current?**	It decreases it.	☐
22 ★	**Explain why step-up transformers are used before electricity is transmitted.**	A higher potential difference means a smaller current, so less energy is wasted heating the cables.	☐
23 ★	**What is direct current (d.c.)?**	A current that flows in one direction only.	☐
24	**Name a source of direct current.**	A cell or battery	☐
25	**State the equation linking power, potential difference and current.**	power = potential difference × current ($P = V\ I$)	☐
26	**State the equation linking power, current and resistance.**	power = current^2 × resistance ($P = I^2\ R$)	☐
27	**State the equation linking energy transferred, power and time.**	energy transferred = power × time ($E = P\ t$)	☐
28	**State the equation linking energy transferred, charge flow and potential difference.**	energy transferred = charge flow × potential difference ($E = Q\ V$)	☐

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	QUESTION	ANSWER	✓
29	**State the equation for a transformer that is 100% efficient.**	p.d. × current in the secondary = p.d. × current in the primary ($V_s \times I_s = V_p \times I_p$)	☐
30	**A battery-powered fan is switched on. Describe the useful energy transfer.**	Energy is transferred electrically from the chemical store of the battery to the kinetic store of the fan.	☐
31 ★	**Explain why step-down transformers are used before electricity reaches homes.**	To reduce the potential difference to a safer value (230 V).	☐
32	**Explain why any connection between the live wire and earth is dangerous.**	A very large current would flow, which could cause a fire or a fatal shock.	☐

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